SYSTEMS BRIEF

Autonomous control for closed living systems

One Bayesian engine reads the true state of a living loop from cheap, drifting sensors, forecasts failure hours ahead, and acts on it autonomously, on-site and offline.

FROM THE MUD TO THE MOON

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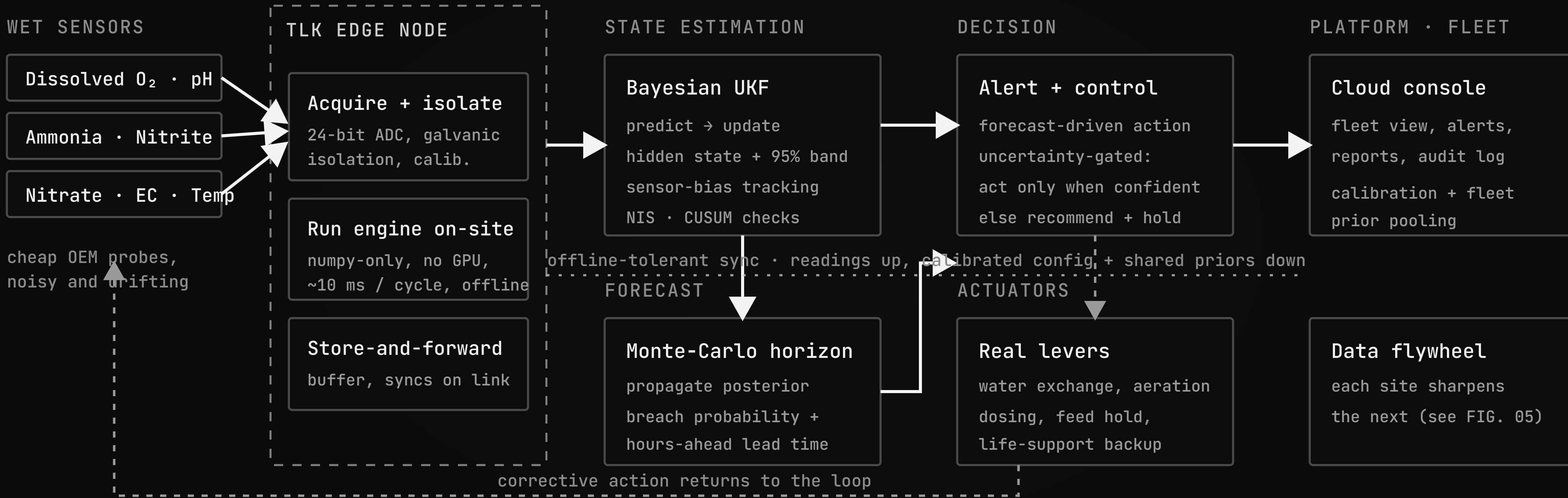
FIG. 01 / SYSTEM

SIGNAL TO ACTION / THE PIPELINE

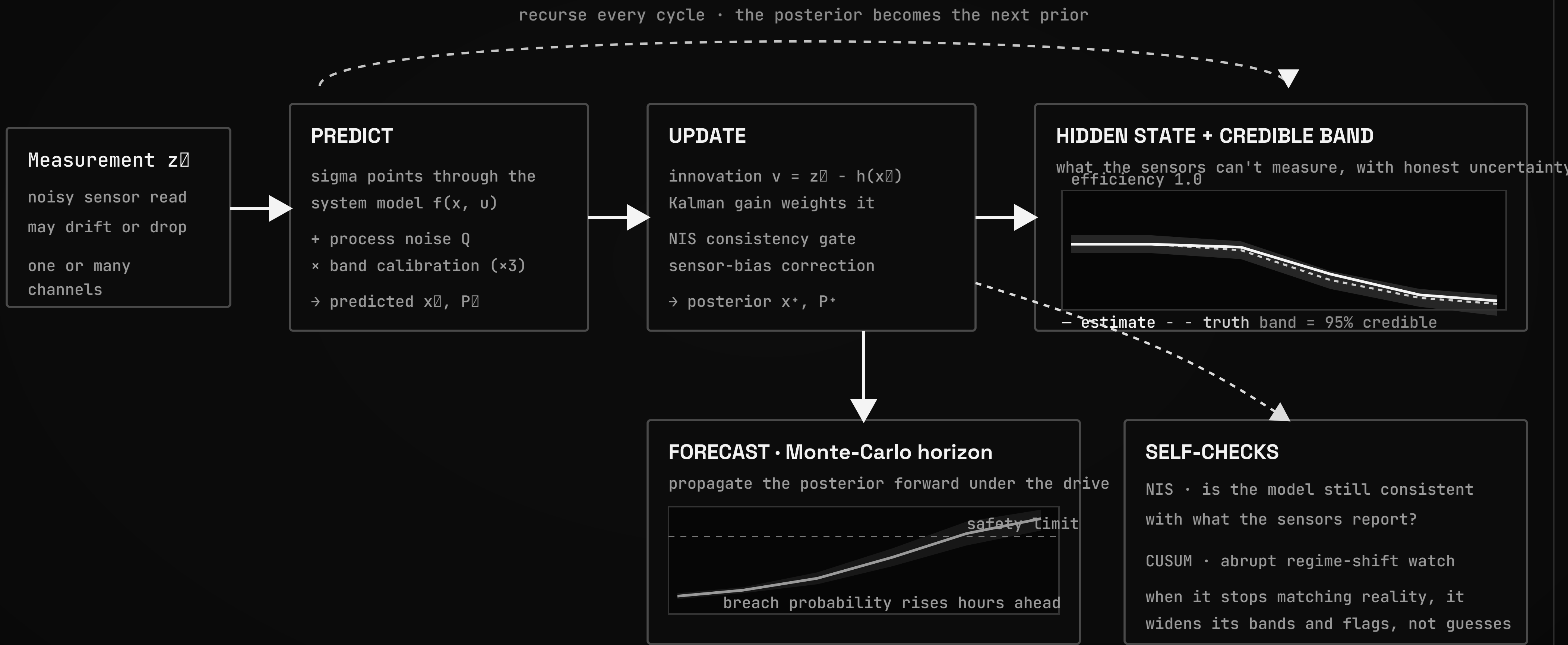
sensor / state data
control / action
store-and-forward sync (offline-tolerant)



Perception and action run at the edge, offline. The cloud coordinates the fleet and closes the learning loop; it is never in the safety path.

One engine drives every domain · aquaculture, soil, bioremediation, and bioregenerative life support · by swapping the system model.

INSIDE THE ENGINE / BAYESIAN STATE ESTIMATION



It infers what it cannot measure from cheap, drifting sensors, quantifies its own uncertainty, and forecasts the breach before it happens.

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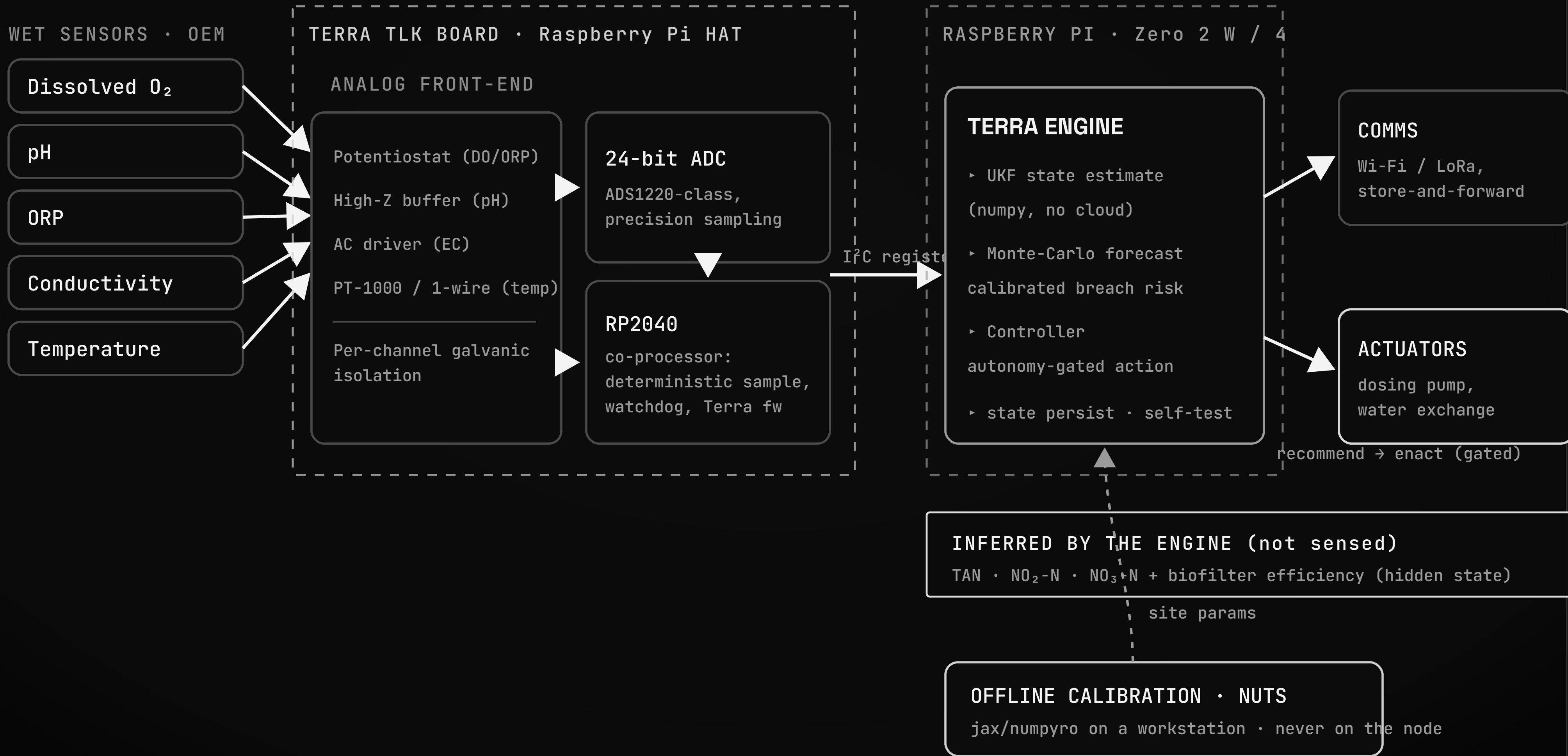
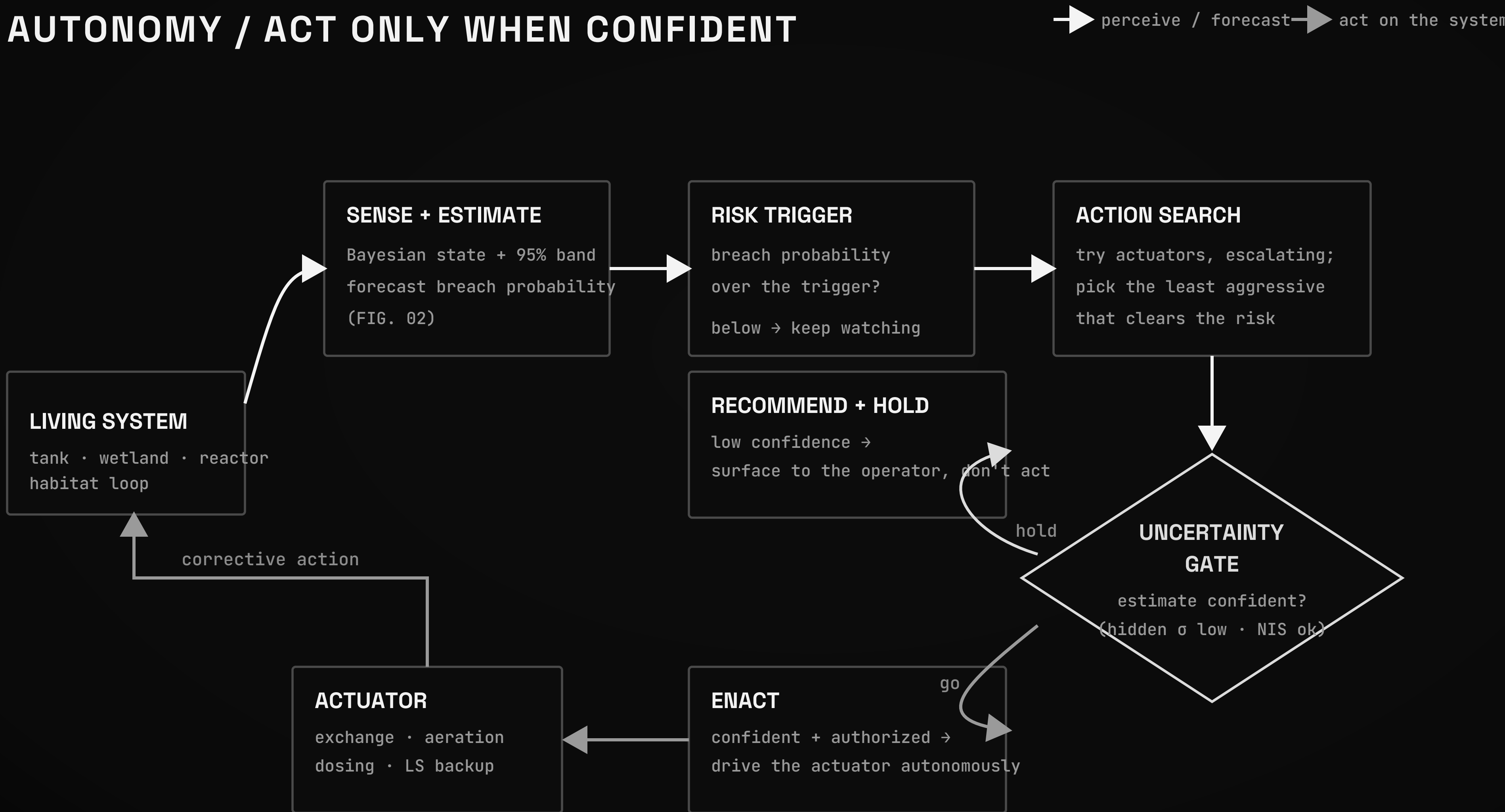
FIG. 03 / EDGE HARDWARE

TLK NODE / SIGNAL CHAIN

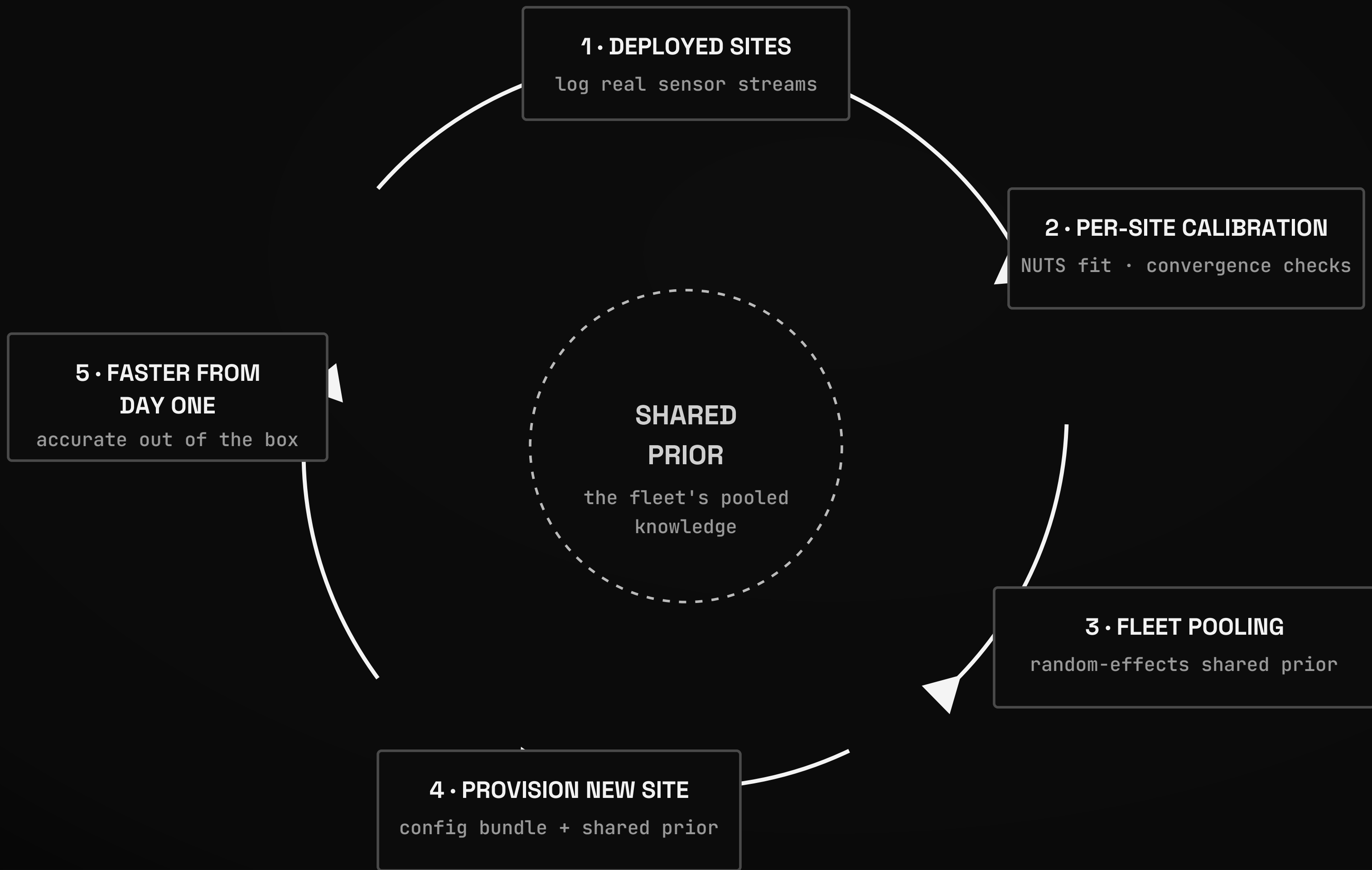


AUTONOMY / ACT ONLY WHEN CONFIDENT

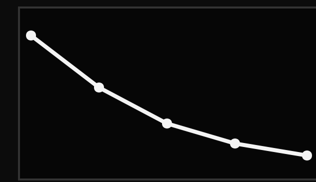


A forecast is only worth what you do with it. The engine acts unattended only when it trusts its own estimate; otherwise it hands the call to a human.

THE DATA FLYWHEEL / EACH SITE SHARPENS THE NEXT



POOLED-MEAN ? vs SITES



2 → N sites: tightens ~ τ / √N

WHY IT COMPOUNDS

the moat is not a patent, it is the outcome-labeled failure data only Terra accumulates in the field

ONE ENGINE / MANY WORLDS

TERRA ENGINE

one Bayesian core, unchanged across domains

- state estimation
- Monte-Carlo forecast
- uncertainty-gated control
- offline + fleet calibration

a new world is a specification, not a new product — swap the system model + sensor map

AQUACULTURE · RAS

hidden state
biofilter nitrification efficiency

sensors
DO · pH · TAN · nitrite · nitrate

prevents
an ammonia / nitrite spike that wipes a stock out overnight

status: engine validated

SOIL · CEA

hidden state
plant-available nutrient pool

sensors
EC · moisture · pH · nitrate

prevents
a nutrient crash or lockout before it stunts the crop

status: engine validated

BIOREMEDIATION

hidden state
microbial activity · electron donor

sensors
ORP · DO · pH · conductivity

prevents
contaminant breakthrough when the culture quietly stalls

status: engine validated

BLSS · LIFE SUPPORT

hidden state
crop photosynthetic capacity

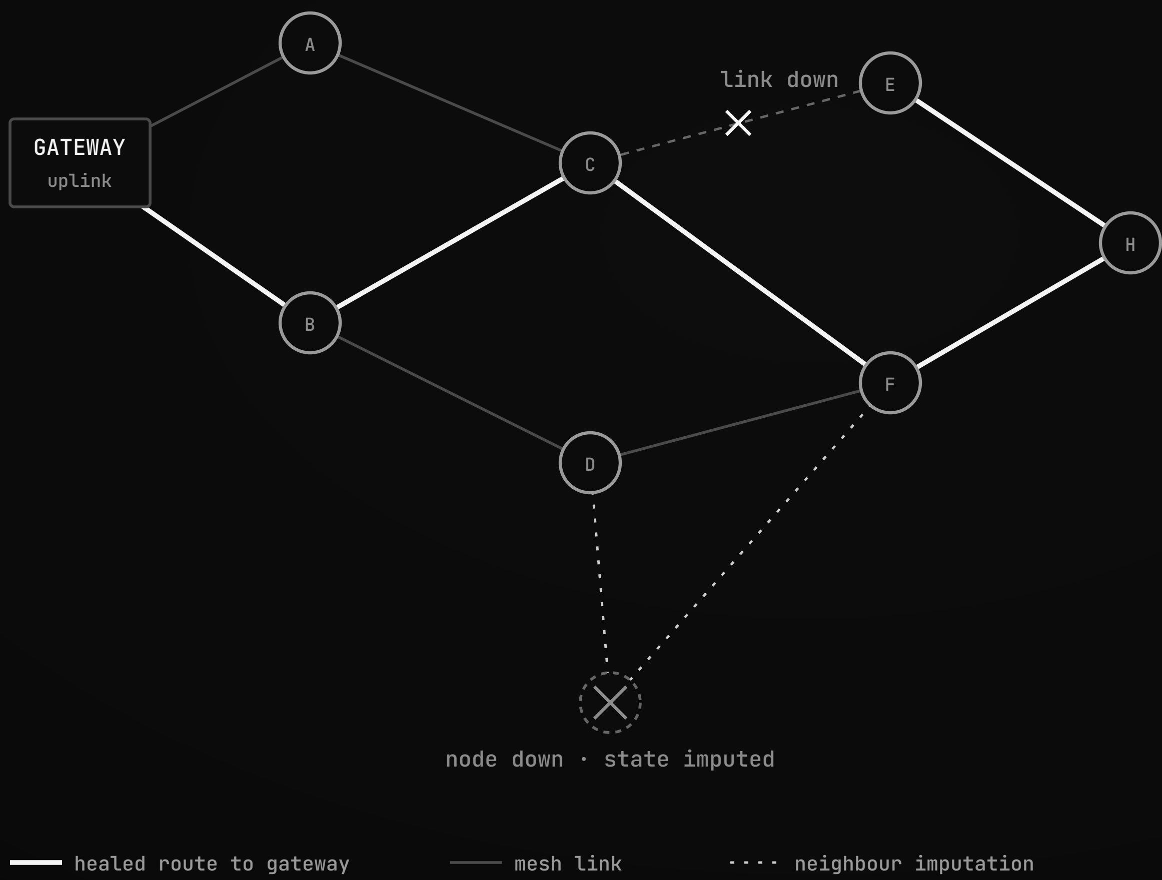
sensors
CO_2 · O_2 · humidity · temp

prevents
a cabin O_2 / CO_2 breach on a long-duration mission

status: NASA-named unmet need

The engine, the offline edge node, and the fleet flywheel are identical everywhere. Only the model of the world underneath them changes.

SELF-HEALING SENSOR MESH



TLK-MESH STACK

RF
sub-GHz LoRa or 2.4 GHz 802.15.4

ROUTING
least-cost to gateway, primary + backup next-hop

HEAL
local repair on missed beacons, no central controller

SYNC
gossip time-sync, coordinated sampling

BUFFER
store-and-forward when partitioned

SECURITY
per-node keys, authenticated frames

SENSING
engine imputes a dropped node's state from correlated neighbours